FLOWSERVE

Thursday Dec-13-2018

From:
Flowserve
1511 Jefferson St.

Sulphur Springs,TX,USA
75482
903-439-3358
kenhorne@flowserve.com

To:

CERTIFICATE OF CONFORMANCE

Serial #	Heat # / Identifier	Customer PO #	Description	Class	End Connection	Shell Test	Seat Test
61-618460	1347S	STP46777	4" 2345	900	FLG	3375	2475
61-618461	1347S	STP46777	4" 2345	900	FLG	3375	2475
61-618462	1348S	STP46777	4" 2345	900	FLG	3375	2475
61-618463	1348S	STP46777	4" 2345	900	FLG	3375	2475

We hereby certify that all the subject valves/parts/sealant have been manufactured in conformance with the applicable requirements of Flowserve Sulphur Springs Operation (FSSO) specifications and that the valve/parts/sealant materials for component parts meet the applicable requirements of FSSO drawings and ASTM specifications, API Q1 9th edition, and API 6D 24th edition (if applicable to the API monogram). All valves/parts/sealant meet all the requirements of the customer purchase order.

Quality Engineer
Kenny Horne

Kenny Horne

FLS-L4-QMSF-QD-009/R1

	FLOW LINK SYSTEMS (P) LTD FOUNDRY UNIT - II 65 / 2-A.B & C,PUDUPALAYAM B.P.O AVINASHI Tk, COIMBATORE - 641 654 INDIA	**TEST CERTIFICATE** (EN 10204-3.1)	TC NO : R-4019 DATE : 3-Nov-18

CUSTOMER	SPECIFICATION REFERENCE	HEAT NO.
FLOWSERVE INDIA CONTROLS PVT LTD, B8, MMDA Industrial Area, Maraimalai Nagar-603209 Kancheepuram, INDIA	1. ASTM A216-18 Gr.WCC 2. Customer Spec. RMC 90012 Rev. 13	1348S Date of Pouring 10-Jul-18 FOUNDRY MARK FLS

PO NUMBER & DATE	ORDER ACCEPTANCE NO
10082358 DT: 15.05.2018	18-0773

CHEMICAL COMPOSITION % wt

MELTING PRACTICE : Steel melted using Electric Induction furnace, and fully killed using Aluminium.

SPECIFICATION	C	Si	S	P	Mn	Ni	Cr	Mo	V	Cu	Al	--	*RE	**CE
Minimum	--	--	--	--	--	--	--	--	--	--	--	--	--	--
Maximum	0.23	0.60	0.020	0.025	1.20	0.50	0.50	0.20	0.030	0.30	0.08	--	1.00	0.43
Achieved	0.202	0.423	0.008	0.015	0.914	0.076	0.068	0.008	0.002	0.012	0.046	--	0.17	0.38

Heat Treatment : NORMALIZING: Temperature raised to 920°C, Soaked for 4 hours and then atmospheric air-cooled

Visual Inspection - Castings conforms to MSS-SP-55-2011 - Type II to XII (a)

MECHANICAL PROPERTIES

TEST TYPE	Tensile as per ASTM A370-17 / ~~ISO 6892-1-2016~~						Impact (CVN) ISO 148-2016/ASTM A370-17		BEND TEST		HARDNESS
SPECIFICATION	YS Rp 0.2% (KSI)	YS Rp 1.0% (KSI)	UTS (KSI)	% Elongation 4d	5d	% RA	Test Temperature (at °C) J Values	Avg.	ANGLE	D	(HBW / HRBW)
Minimum	40	--	70	22	--	35	--	--	--	--	--
Maximum	--	--	95	--	--	--	--	--	--	--	--
Achieved	43.7	--	74.8	33	--	68	--	--	--	--	147

DETAIL OF CASTINGS

S.No.	PO S.No.	DESCRIPTION	CAST DRAWING NO / REV	PART NO	POURED QTY	DESP. QTY
1	01	4" 900 FE BODY	D-487876 / S	N2757928-60	7	1

REMARKS:

1 Stated Chemical composition is from heat analysis; and Mechanical properties of test coupon from Cast keel block

2. *RE = Ni + Cr + Mo +V +Cu

3 ** $CE = C + \frac{Mn}{6} + \frac{Cr+Mo+V}{5} + \frac{Ni+Cu}{15}$

4 The above stated KSI values have been calculated based on original tested N/MM² values.

MTC - REVIEWED & ACCEPTED
10139336 FICPL QC

CERTIFIED THAT
1. THE ABOVE GIVEN DETAILS ARE CORRECT, FULFILLS THE SPECIFICATION / STANDARDS AND PO.
2. MATERIAL WAS MANUFACTURED, SAMPLED, TESTED, INSPECTED IN ACCORDANCE WITH THE ABOVE SPECIFICATIONS AND CERTIFIED AS PER EN 10204 - 3.1.
3. THE MATERIAL IS FREE OF CONTAMINATION FROM RADIO ACTIVE ELEMENT OR RADIATION.

PREPARED BY:	CHECKED BY:	
S.MANIBHARATHI	R.SATHISH KUMAR ASSISTANT ENGINEER	M.S.DHANABALAN ASSISTANT MANAGER -QUALITY MANUFACTURER'S AUTHORIZED INSPECTOR